Dark Matter Behaves Like Matter: A Halo Rule Set by Starlight Predicts Dwarf Galaxies Better Than MOND and Two Modified Gravity Theories

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Abstract

One formula for the missing mass of galaxies, with five numbers fitted on disk galaxies and one threshold set on published lensing masses, predicts the accelerations of 310 galaxies and clusters, across nine decades in stellar mass, to within their measurement errors: held out, its error on every dynamical scale, from dwarf spheroidals to clusters, lies within the range that the published errors alone produce. It matches or beats both modified Newtonian dynamics (MOND) and standard abundance-matched dark-matter halos at every scale tested, and over all rotation-curve points, with or without the standard cut on their quality. In seven tests against measurement noise, the six dynamical scales and the lensing on the stricter of two preset masks, its worst result is one that a perfect formula would show one time in five, while MOND's relation fails five and standard halos fail the lensing. The missing mass behaves like matter because its amount follows the stars. The formula's backbone, a three-number halo rule driven by stellar mass and size, predicts unseen spirals as well as MOND's acceleration relation (0.1533 against 0.1531 dex) and, unchanged, 44 dwarf spheroidals 28 percent better (0.479 against 0.669 dex) and 25 more outside the original sample 45 percent better. Over 85 dwarfs with upper limits, it beats MOND's relation by 36 log-likelihood units. Joined with the published Moster halo relation through cold gas and each sample's selection, with clusters' hot gas counted, the rule cuts

cluster errors from 0.77 to 0.10 dex, though blind it predicts only typical cluster masses. The last two numbers give the halos of diffuse gas-rich disks a core, which cuts the error on the rotation-curve points that the benchmark's cut on velocity errors removes by 14 percent, to 5 percent below MOND's relation held out (2 percent if MOND's constant is refitted the same way), and below it on 34 galaxies outside the benchmark. With its one number set on the same disks and carried to every halo, none of fuzzy, warm and self-interacting dark matter passes the three tests that the core passes: on this benchmark the cores follow the gas, not the halo. With published constants, MOG and nonlocal gravity strengthen gravity at a half-mass radius at most 19 and 23 times, too little for most dwarfs, and MOND's hot-gas cluster repair, carried to gas-rich spirals, raises speed errors from 12 to 100 percent. Frozen on rotation curves, the rule predicts the amplitude of Kilo-Degree Survey lensing within 0.05 dex per stellar-mass bin but not its profile; five patterns within the formula's residuals exceed the noise; and ten hashed forecasts name the data that would test each open point.

Keywords: dark matter; modified gravity; MOND; radial acceleration relation; dwarf spheroidal galaxies; galaxy clusters; nonlocal gravity; MOG; stellar-to-halo mass relation; weak lensing; initial mass function; halo concentration; self-interacting dark matter; cross-validation; blind test; forecasts

1. The question and the claim

One formula for the missing mass of galaxies, fitted only on disk galaxies and published data, matches or beats both MOND and standard dark-matter halos at every scale tested here, from dwarf galaxies to clusters and lensing (Section 6.3), and over all rotation-curve points, with or without the usual cut on their quality (Section 6.4). The missing mass of galaxies behaves like matter because its amount follows a galaxy's stars. A halo predicted from stellar mass and size

forecasts galaxies it has not seen, while the laws that change gravity tested here fail when carried from one kind of system to the next.

Galaxies rotate too fast for their visible matter. Two kinds of answer compete. One adds matter: each galaxy sits in a halo of dark matter, and the halo's mass and concentration follow from how structure formed (Navarro, Frenk and White, 1997). The other changes the law: gravity strengthens below an acceleration scale a0, as in modified Newtonian dynamics, MOND, whose radial acceleration relation ties the observed acceleration in disk galaxies to the acceleration of their ordinary matter alone (McGaugh, Lelli and Schombert, 2016; Lelli et al., 2017).

On disk galaxies both answers fit well, which is why the question has stayed open. The radial acceleration relation needs one constant, and halo fits need two per galaxy. A fair test needs what a fit to one population cannot give: predictions for systems the model has not seen, from measured inputs only, carried across populations whose sizes, gas content and histories differ.

This paper makes that test across 286 systems, at radii from 27 parsecs in dwarf galaxies to 1,000 kiloparsecs in clusters, and adds the stacked lensing of 259,383 galaxies. Comparisons usually fit the galaxies they score; here every rule is fixed on some systems and scored on others, and twenty-one written protocols, each hashed before its own scoring, govern the tests that followed the first comparisons (Table 1). Section 3 sets a halo rule with three shared coefficients on disk galaxies and carries it, frozen, to dwarfs and to lensing. Section 4 gives a bound that explains why modified gravity that copies visible mass by a fixed recipe, MOG and nonlocal gravity included, falls short on most dwarfs. Section 5 shows that one acceleration scale, and the 2025 repair of MOND in clusters, fail when carried from one population to the next, while a published halo relation set by starlight carries to cluster scales. Section 6 joins that relation and the rule in one formula, scores it held out against MOND's relation, standard halos and the

noise of the measurements, puts three kinds of dark-matter particle in the place of its core, and lists what remains with ten hashed forecasts. Section 7 is a warning: cluster benchmarks scored at R_500 are circular, since that radius is defined by the mass they test. Supplemental Notes S1 to S12 give each test in full, with its sensitivity checks, audits and failed attempts.

2. Data and the rules of the test

2.1 Five populations

The main sample holds 279 systems in five populations, and seven more clusters serve as a separate stress test.

Rotating disk galaxies come from SPARC (Lelli, McGaugh and Schombert, 2016): 175 galaxies and 3,391 radial measurements. Selecting quality flag below 3, inclination at least 30 degrees, positive radius and speed, relative speed error at most 0.1 and positive inward baryonic acceleration leaves 149 galaxies and 2,700 measurements. The measurements with larger speed errors, 582 over all 175 galaxies, are examined separately (Section 6.4). Disk and bulge mass-to-light ratios stay fixed at 0.5 and 0.7 in solar units at 3.6 micrometers, with catalog distances and inclinations. The baryonic speed is V_bar^2 = V_gas|V_gas| + 0.5 V_disk|V_disk| + 0.7 V_bulge|V_bulge|, and the accelerations are g_bar = V_bar^2/r and g_obs = V_obs^2/r.

Dwarf spheroidal galaxies come from McConnachie's catalog of nearby galaxies in its January 2021 update (McConnachie, 2012): 25 classical and 19 ultrafaint systems, divided at a V-band luminosity of 10^5 suns. Each gives one point at the half-light radius in three dimensions, r_1/2 = (4/3) R_e, with g_obs = 3σ^2/r_1/2 from the line-of-sight dispersion σ (Wolf et al., 2010) and g_bar = G(M_star/2)/r_1/2^2, the stellar mass set at twice the V-band luminosity.

Mass-to-light ratios of one and three are kept as sensitivities. Unresolved dispersions, flagged tidal systems and measured gas above a tenth of the stellar mass are excluded.

Sixteen early-type galaxies give one inner Jeans estimate and one outer neutral-hydrogen estimate each (Lelli et al., 2017), with a photometric mass-to-light ratio of 0.8 at 3.6 micrometers. Fifty brightest cluster galaxies from MaNGA give isotropic Jeans estimates (Tian et al., 2024). Twenty CLASH clusters give 84 lensing estimates, at their centers and at 100, 200, 400 and 600 kiloparsecs (Tian et al., 2020). Seven X-COP clusters give 21 hydrostatic estimates at fixed radii of 500, 750 and 1,000 kiloparsecs (Ettori et al., 2019; Eckert et al., 2022).

The accelerations for dwarfs, early-type galaxies, cluster centers and clusters are inferences, each carrying its source's dynamical or lensing model. They are treated as data with those assumptions attached, and no score below pools populations into one ranking.

2.2 Held out by system

Every score is a prediction for systems left out of the fit. Folds hold out whole galaxies or whole clusters, all radii of a system stay in one fold, and objects that may share a host stay together. Population tests go further: a rule fitted on one population is frozen and applied to another, with no target-population calibration. Coefficients are refitted inside each training split, each galaxy weighs equally in fits and scores, and the reported score averages three complete assignments of galaxies to folds.

Scores are root-mean-square errors in log10 acceleration, in dex. An error of 0.15 dex is a factor of 1.41; 0.48 dex is a factor of 3. Speed errors take the square root of the acceleration ratio, so a 0.15-dex acceleration error is 0.075 dex in speed,

about 16 to 19 percent. The median speed errors quoted below are a different statistic, the median absolute fractional error.

The data had been inspected before these comparisons, and each protocol was fixed before its own run. The results on the main sample are exploratory comparisons of declared rules, not a preregistered discovery. The later tests ran under twenty-one written protocols, each hashed before its own scoring, on data inspected in earlier runs; Table 1 lists them with their outcomes, those that failed included. A hash fixes a file's content, not when it was written; the times given are the author's records.

Table 1. The twenty-one protocols, each written and hashed before its own scoring: what each fixed, and how it came out. Notes S1 to S12 give each in full.

Protocol	Section	What it fixed before the run	Outcome
1, 2	3.3	25 dwarf spheroidals outside the original 44, every rule frozen on the disks; 17 of them scored blind	Stellar rule 0.441 dex against 0.808 for MOND's relation; the range of the gain excludes zero
3	3.4	Five dwarfs with measured dispersions and ten with upper limits that no rule had scored	Lowest error of seven models on the five, not significant; two limits exceeded where MOND meets all ten. Neither preset test met
4	3.5, 3.6	One likelihood for all 85 dwarfs; forecasts for 264 dwarfs not yet measured	Ahead of MOND's relation by 36 units (24 to 48). Test met
5	5.3	The published Moster relation, nothing fitted, on 50 MaNGA galaxies and 20 CLASH clusters	0.185 dex against 0.403 for MOND's relation. Test met
6	5.3	Audit: the same relation inverted with its scatter and the halo mass function	0.288 dex against 0.403
7	5.3	Audit: the Moster halo masses against published lensing masses	Masses not recovered: they scatter by 0.73 dex
8	3.8	Lensing halo masses of SDSS central galaxies	Blue centrals within 0.16 dex in four of five bins, 0.34 dex high in one. Test not met

Protocol	Section	What it fixed before the run	Outcome
9	3.8	Stacked KiDS lensing of 259,383 isolated galaxies	Amplitude within 0.05 dex in every stellar-mass bin; blue galaxies 0.14 dex low; profile not matched
10 to 16	6.1, 6.2	The one formula, built in steps and scored on 60 held-out partitions; 12, blind on 216 HeCS clusters; 15, threshold set on SDSS red centrals; 16, clusters' hot gas counted	Table 7. Cluster error cut from 0.77 to 0.10 dex; blind, only a sample's typical cluster mass is recovered
17, 18	6.4	Two terms for gas-rich disks: a fitted weight on cold gas; larger, looser halos	Both failed their marks
19	6.1, 6.4	A core for the halos of diffuse gas-rich disks, with three marks	All three met
20	6.5	Fuzzy, warm and self-interacting dark matter, each in the core's place, with the core's marks	None meets them
21	6.5	A gate for the core built on the history of star formation	Not decided by the data in hand

3. Halos set by starlight

3.1 The rule

Each galaxy gets a finite NFW halo, density ρ_s/[z(1+z)^2] with z = r/r_s, truncated at the radius r_200 enclosing 200 times the critical density. Its concentration c_200 = r_200/r_s follows the published Planck relation of Dutton and Macciò (2014),

c_200 = 10^0.905 (M_ref / 10^12 h^-1 M_sun)^-0.101, h = 0.671,

so only the halo's reference mass M_ref needs predicting. The rule predicts it from the galaxy's stellar mass M_star = 0.5 L_3.6 and its effective radius R_e:

M_ref / M_sun = M_p/(f_b M_sun) + 10^A (M_star/10^10 M_sun)^B (R_e/3 kpc)^C,

where M_p = M_star + 1.33 M_HI is the galaxy's baryonic mass and f_b = 0.0490/0.3175 is the cosmic baryon fraction. The first term keeps the halo at least as massive as the baryons allow. The halo's dark mass is (1 - f_b) M_ref, and its pull adds to the measured baryonic pull. Fitted to all 149 disks, A = 11.748, B = 0.818 and C = -0.712; every prediction below uses coefficients fitted without the system predicted.

No rotation measurement of a predicted galaxy enters its prediction: stellar luminosity, gas mass and size set the halo, and the halo and the baryons set the curve. Stellar-to-halo relations have long precedent (Moster, Naab and White, 2013), and baryon-halo scaling has been tied to the radial acceleration relation before (Navarro et al., 2017; Li et al., 2019). What is tested here is whether such a rule, with three coefficients, carries from disks to systems it never saw. B and C nearly cancel (B + C = 0.11), so the rule depends mainly on how compact the stars are, M_star/R_e, a measure of the depth of the stellar potential. One reading is that a deeper halo pulls cooling gas into a more compact galaxy; this paper does not test it.

3.2 Disks: as good as MOND's relation

Table 2. Shared predictors fitted on the disks: error on unseen disks and, frozen, on the 44 dwarfs, in dex.

Shared predictor	Coefficients	Unseen disks, dex	Frozen dwarfs, dex
Radial acceleration relation, refitted a0	1	0.15309	0.66937
Halo set by baryonic mass	3	0.15323	0.66757
Published Moster relation, no fitting	0	0.17111	0.55591
Halo set by stellar mass and size	3	0.15329	0.47914
Curved stellar relation	5	0.15380	0.47874

Fitted on all the disks with each galaxy weighted equally, the relation takes a0 = 1.04 × 10^-10 m/s^2 and the baryonic rule A = 11.467, B = 0.994 and C = -0.635. On unseen disks the stellar rule matches the radial acceleration relation to 0.0002 dex, and its median speed error is 9.09 percent. Two predictors built on opposite assumptions describe disk rotation equally well from the same inputs, which is why disks alone cannot settle the question. A curved stellar relation with five coefficients adds nothing.

3.3 Dwarfs: 28 percent better

Frozen on the disks and applied to the 44 dwarf spheroidals, using only their stellar mass and size, the stellar rule cuts the error from 0.669 dex for the radial acceleration relation to 0.479 dex, 28 percent lower. A paired bootstrap over whole dwarfs puts the gain over the baryonic-mass halo at -0.188 dex, with a 95 percent interval of -0.281 to -0.083. Thirty of the 44 dwarfs improve. The relation does no better when its own constant is refitted on the dwarfs and scored on dwarfs held out: 0.548 dex (Section 4.3). The stellar rule was proposed after the baryonic rule's dwarf transfer had been seen; its coefficients come from the disks alone.

The transfer holds on a second compilation and on dwarfs outside the original sample (Note S1). On a fixed version of the Local Volume Database (Pace, 2025), which gives each measurement with its source, 42 dwarf spheroidals with measured dispersions, 33 of them among the 44, give the same ranking: 0.389 dex for the stellar rule against 0.611 for the radial acceleration relation. The 25 dwarf spheroidals in the same database that are not among the 44 and pass the same selection were then scored with the rules frozen on the disks, 17 of them blind, under a protocol fixed and hashed before any was seen. The stellar rule scores 0.441 dex against 0.808 for the radial acceleration relation, 45 percent

lower; the paired bootstrap interval of its gain over the relation, -0.449 to -0.272 dex, excludes zero, and 21 of the 25 dwarfs improve. The published Moster relation, a second halo set by starlight and fitted to no rotation data, also beats MOND's relation on all three dwarf sets: 0.403 dex on these dwarfs, and 0.556 and 0.463 on the original 44 and the second compilation. A learning curve shows that the transfer is learned from the disks, not tuned to any of them: refitted on random subsets of the disks, the stellar rule beats the relation on both dwarf sets in every one of 50 draws from 30 training disks up.

MOND's defenders answer that a satellite's own gravity is weakened by its host's field, the external field effect, which Chae et al. (2021) report in the rotation curves of disk galaxies. Applied as McGaugh and Milgrom (2013) apply it to Andromeda's dwarfs, it lowers MOND's predictions for dwarfs that already move faster than MOND predicts: its error rises to 0.96 dex on the original 44 and 1.20 on the 25 out of sample. On the 25 brighter dwarfs of the original sample it matches the stellar rule, 0.41 against 0.40 dex; on the 19 ultrafaint ones it misses by 1.39 dex. MOND's answer for those, that they are out of equilibrium, needs evidence that one point per dwarf cannot give.

The halo has an environment too. In simulations, a subhalo's concentration rises toward its host's center (Moliné et al., 2017). With the published coefficient, that factor lowers the stellar rule's error to 0.457 dex on the original 44 and 0.383 on the 25 out of sample. It was tried after the residuals were seen to grow toward the host, and the next blind test rejected it (Section 3.4), so the rule leaves it out; the one formula of Section 6.1 carries it for satellite dwarfs, where with a floor on halo mass it improves the held-out dwarfs by 0.036 dex. For 31 Milky Way satellites with measured pericenters (Pace, Erkal and Li, 2022), the rule's residuals show no trend with pericenter or present distance at fixed luminosity (Note S1).

About half of what remains is measurement. The quoted errors of the dispersions and radii give 0.30 dex on the original 44 and 0.34 dex on the 25; a likelihood that separates them leaves an intrinsic scatter of 0.32 and 0.33 dex, about a factor of 2, for differences between halos and for orbits, tides and binary stars, which a rule given only stellar mass and size cannot see.

The one change between the two halo rules is which mass sets the halo: stars or all baryons. Disks cannot tell them apart. Dwarfs, which have stars and almost no gas, can. The halo follows the stars.

The gain is not uniform. Ultrafaint dwarfs improve from 0.934 to 0.571 dex, and classical dwarfs worsen from 0.348 to 0.395. Satellites of the Milky Way improve, and those of Andromeda worsen. Three of four tails of the disk sample, cut by predictor values, worsen slightly. The remaining dwarf errors, a factor of about 3, are large, and satellite evolution, tides and binary stars are not modeled. The result is a better transfer, not a solved dwarf problem.

3.4 Dwarfs no rule had scored

The database holds two kinds of dwarfs that no earlier test scored. Some lie beyond the Milky Way and Andromeda systems, and some have only an upper limit on their dispersion. A third protocol, written and hashed before any prediction for these dwarfs was computed, scored both kinds with every rule frozen (Note S2). Five dwarfs have measured dispersions, four of them far from any large galaxy, where neither of MOND's answers for satellites applies. On the five the stellar rule has the lowest error of seven models, 0.171 dex against 0.202 for MOND's relation, but five dwarfs cannot make that significant: the bootstrap interval of the gain runs from -0.170 to +0.144 dex, and the ranking changes with major-axis radii or a mass-to-light ratio of 3. The protocol's first test is not met. Ten satellites have only upper limits. The stellar rule's predicted dispersion lies

below eight of them and exceeds the limits of Grus II (Simon et al., 2020) and Andromeda XII (Collins et al., 2013); MOND meets all ten, so the second test is not met either. MOND passes because it predicts slow motion for every faint dwarf, and the same predictions miss the faint dwarfs with measured dispersions in Section 3.3 by factors above 20 in acceleration. Upper limits can catch only a rule that predicts too much motion; measured dispersions catch errors both ways. Section 3.5 weighs both kinds in one score.

3.5 All 85 dwarfs in one score

Measured dispersions and upper limits can be weighed together. A fourth protocol, hashed before any likelihood was computed, puts all 85 dwarfs scored so far into one likelihood, every value taken from the database: 75 measured dispersions count as values with their errors, and 10 upper limits count as ceilings. Each model keeps its prediction as the center, and its one free number is its scatter (Table 3).

Table 3. All 85 dwarfs in one likelihood, 75 measured and 10 upper limits, every rule frozen. Each model's only free number is its scatter, in dex. The last column is how far its log-likelihood falls below the stellar rule's.

Model	Scatter, dex	Log-likelihood	Behind stellar rule
Halo set by stellar mass and size	0.30	-39.3	0
Stellar rule with subhalo factor	0.29	-36.9	-2.4
Published Moster relation	0.40	-54.4	15.1
Radial acceleration relation, published a0	0.56	-72.8	33.5
Radial acceleration relation, disk a0	0.58	-75.4	36.2
Halo set by baryonic mass	0.58	-75.7	36.4
MOND with external field	0.93	-106.0	66.7

Among the baseline frozen rules, the stellar rule fits best, with a scatter of 0.30 dex; the exploratory subhalo variant, discussed below, scores slightly higher. It beats MOND's relation by 36 units of log-likelihood, with a bootstrap interval of 24 to 48 units, which meets the protocol's test. It beats MOND with the external field by 67 units. The limits cost the stellar rule 4 units more than MOND's relation, and the measured dwarfs give it 40 units more. On average the stellar rule sits 0.02 dex from the data, while MOND's relation sits 0.43 dex too low. Even when MOND's relation may shift its whole prediction to fit the dwarfs, the unshifted stellar rule stays ahead by 9 units.

The gain over MOND's relation stays between 27 and 48 units for mass-to-light ratios of 1 and 3, for major-axis radii, without measurement errors, and with every measured dispersion below 3 km/s dropped, a check on binary stars. With each dwarf held out in turn and each model's scatter fitted to the other 84, the stellar rule leads MOND's relation by 35.7 units (21.8 to 47.8), and 70 of the 85 dwarfs favor it. The subhalo factor scores 2.4 units better than the rule without it, inside its bootstrap interval of -8.6 to +3.3, so its net effect is not established (Note S3).

3.6 Forecasts for dwarfs not yet measured

The same protocol fixed the dispersion each rule forecasts for the 264 gas-poor dwarfs in the database with no measured dispersion. The file was hashed on 31 October 2026, and its SHA-256 begins bbafc29918f89053. Table 4 lists the nine in the Local Group, with 68 percent ranges from the scatter of Section 3.5. For these, the stellar rule forecasts dispersions a median 1.5 times MOND's with the external field. The protocol calls a forecast decisive when the two differ by more than a factor of 2, which among the nine holds for Carina IV alone, the faintest: 2.1 km/s from the stellar rule against 0.84 from MOND. The other 255 lie farther out, most in groups of galaxies up to 13 megaparsecs away, where the two

forecasts differ by a median factor of 1.3. The ranges are a factor of 1.42 either way for the stellar rule and 2.9 for MOND, whose scatter must be wide to cover its misses. Measured together, the nine separate the two forecasts by 1.4 standard deviations, so the set weighs more than any single dwarf. Measured dispersions near MOND's for the nine would count against the rule, and near the rule's, against MOND.

Table 4. Forecast line-of-sight dispersions, km/s, with 68 percent ranges, for the Local Group dwarfs with none measured, faintest first. MOND includes the field of the host, or for dwarfs with no host the stronger of the Milky Way's and Andromeda's fields.

Dwarf	Host	M_V	Stellar rule	MOND	Ratio
Carina IV	Milky Way	-2.8	2.11 (1.49 to 2.99)	0.84 (0.29 to 2.44)	2.50
Leo K	none	-4.73	3.09 (2.18 to 4.38)	1.80 (0.62 to 5.23)	1.71
Andromeda XXXV	M31	-5.2	3.07 (2.17 to 4.35)	2.01 (0.69 to 5.84)	1.53
Pegasus VII	M31	-5.7	4.13 (2.92 to 5.85)	2.25 (0.77 to 6.54)	1.83
Leo M	none	-5.77	3.76 (2.65 to 5.33)	2.29 (0.79 to 6.66)	1.64
Andromeda XXXVI	M31	-5.91	3.49 (2.46 to 4.94)	2.37 (0.82 to 6.89)	1.47
Pegasus V	M31	-6.3	3.91 (2.76 to 5.54)	2.59 (0.89 to 7.53)	1.51
Tucana B	none	-6.9	4.20 (2.97 to 5.95)	2.97 (1.02 to 8.64)	1.41
Pegasus W	none	-7.2	4.50 (3.18 to 6.37)	3.18 (1.09 to 9.25)	1.41

3.7 What does not help

Published feedback profiles, which change halos by the gas that star formation drives out and which fit individual galaxies well (Li et al., 2020), do not improve the shared rule: the DC14 recipe (Di Cintio et al., 2014) gives 0.189 dex on disks, and coreNFW (Read, Agertz and Collins, 2016) with an assumed formation time gives 0.163 dex. That holds on the benchmark's points and for cores given to every

galaxy by a published recipe; on the noisier points that the benchmark's quality cut removes, a core confined to diffuse gas-rich disks is needed, and the final formula carries one (Sections 6.1 and 6.4). A blend of the radial acceleration relation and the stellar halo lowers the disk error by under 2 percent, a gain its uncertainty does not establish, and transfers to dwarfs less well than the stellar halo alone; allowing halo scatter improves a predictive score without improving the central error (Note S4).

The stellar rule is calibrated on single disk galaxies and is not carried to the heavier systems. Above about the Milky Way's stellar mass, the relation between a galaxy's stars and its halo bends, as mergers add halo mass faster than stars (Moster, Naab and White, 2013), and a cluster's potential sums the halos of hundreds of member galaxies and its hot gas. Standard structure formation predicts that change; a universal force law has to hold with one constant through it, and Section 5 tests whether it does. Section 5.3 carries the published Moster relation, which includes the bend, to cluster scales, and Section 6.1 joins it to the rule.

3.8 Whole halos from lensing

Rotation curves reach a few tens of kpc, while galaxy-galaxy lensing weighs whole halos. Two protocols, the eighth and the ninth, each hashed before its own scoring, carry the rule frozen on the disks into lensing (Note S5). Mandelbaum et al. (2016) measured the mean halo masses of red and blue central galaxies in the Sloan survey, in bins of stellar mass. Given the disks' own scalings of size and gas with stellar mass, the rule predicts the halos of the blue, star-forming centrals within 0.16 dex in four of five bins from 10^10.2 to 10^11.3 suns, a median 0.05 dex high. At 10^10.56 suns it is 0.34 dex high, so the protocol's test, every bin within 0.2 dex or twice its error, is not met. Above 10^11 suns the red centrals'

halos grow about as the square of stellar mass and lie above the rule by 0.3 dex at 10^11.1 suns and 1.0 dex at 10^11.7: the bend of Section 3.7, reached by the centrals of groups and clusters.

Lensing also traces how the mass is spread. Brouwer et al. (2021) stacked the Kilo-Degree Survey (KiDS-1000) lensing signal of 259,383 isolated galaxies at redshifts 0.1 to 0.5 in 15 bins of baryonic acceleration, spanning about 30 kpc to 3 Mpc, for four bins of stellar mass and for early and late types of matched stellar mass. Inside 0.3 Mpc, where false isolated lenses add little, the measured signal lies 0.027 ± 0.015 dex below the frozen prediction for the whole sample, and between 0.005 and 0.048 dex below it in each of the four stellar-mass bins from 10^8.5 to 10^11 suns. For blue and disk-dominated galaxies it lies 0.14 and 0.12 dex below, outside the protocol's 0.1 dex. The rule puts early types only 0.05 dex above late types of the same stellar mass, against a measured 0.16 dex by Sérsic index and 0.21 dex by color: about a third and a quarter of the split.

These are amplitudes; the radial profile does not match. With the release's covariance, the whole sample's seven bins inside 0.3 Mpc give a chi-square of 45.1 for six degrees of freedom once the amplitude is fitted ($p = 4 \times 10^{-8}$). The measured signal lies 0.19 dex below the prediction in the innermost bin and 0.16 dex above it in the bin nearest 0.3 Mpc, so the measured profile is flatter than the predicted one, and all 14 of the protocol's sensitivity cases miss it as well. Nor does lensing yet support the rule's size term at fixed stellar mass and type: in CFHTLenS, at fixed stellar mass and color, halo mass grows with size (Charlton et al., 2017), where the rule's halo mass falls as size to the power of about -0.6. A lensing measurement of halo mass against size at fixed stellar mass and type, for central galaxies and with the rule's prediction fixed first, would decide it.

4. A mass budget no fixed kernel can meet

4.1 The bound

In the weak-field limit, nonlocal gravity (Rahvar and Mashhoon, 2014) and MOG with constant parameters (Moffat and Rahvar, 2013) both act as if the ordinary matter carried extra effective matter, spread from it by a fixed positive kernel q:

$\nabla^2\Phi = 4\pi G(\rho_b + \rho_{eff})$, $\rho_{eff}(x) = \int q(|x - y|)\, \rho_b(y)\, d^3y$,

with $q_{NLG}(d) = \alpha\mu(1 + \mu d)e^{(-\mu d)}/(8\pi d^2)$ and $q_{MOG}(d) = \alpha\mu^2 e^{(-\mu d)}/(4\pi d)$. Each kernel integrates to α over all space. An isolated source of baryonic mass M_b therefore carries total effective mass αM_b, and no part of space can hold more of it than that. At the source's own half-mass radius $r_{1/2}$, inside which the baryonic mass is $M_b/2$, the enclosed effective mass is at most αM_b, so

$g/g_b = 1 + M_{eff}(< r_{1/2})/(M_b/2) \le 1 + 2\alpha$.

The bound holds for any positive spherical source, whatever the kernel's range $1/\mu$ and whatever the source's radial profile. For a source whose density falls outward, such as a Plummer sphere, no displaced ball encloses more mass than the centered one, and the ceiling tightens to $1 + \alpha$, since then $M_{eff}(< r_{1/2}) \le \alpha M_b/2$.

4.2 The dwarfs exceed it

The published fitted values (α, μ in kpc^-1) are (10.94, 0.059) for nonlocal gravity and (8.89, 0.042) for MOG, so the general ceilings are 22.9 and 18.8, and for a Plummer sphere 11.9 and 9.9. The dwarf estimates sit at $(4/3)R_e$, where a Plummer source encloses 0.512 of its mass, so the general ceiling there is $1 + \alpha/0.512$: 22.4 and 18.4. At a stellar mass-to-light ratio of 2, the inferred ratio g_{obs}/g_{bar} of 31 of the 44 dwarfs exceeds the nonlocal gravity ceiling, and 33 exceed MOG's. At a mass-to-light ratio of 3, 29 exceed each. These are counts of

central estimates, and the estimates carry the assumptions of dynamical equilibrium, membership, anisotropy and binary stars; but the bound itself needs no fitting, and most dwarfs sit well above it.

4.3 Integrating the kernels

The bound is a ceiling. The full prediction integrates each kernel against a Plummer sphere fixed by each dwarf's stellar mass and size, with matter outside the half-mass radius included:

Table 5. Each dwarf's stars spread as a Plummer sphere with half its mass inside r_1/2, with the relation and the two kernels integrated over it, in dex. Refitted models are scored on dwarfs held out in five folds, in the first of three assignments; the mean of the three is 0.543 for the relation and 0.722 for the kernels.

Model, mass-to-light ratio 2	Dwarfs held out, dex
Published radial acceleration relation, fixed a0	0.651
Radial acceleration relation, a0 refitted on dwarfs	0.548
Published nonlocal gravity kernel	1.862
Published MOG kernel	1.905
Nonlocal gravity, both coefficients refitted	0.729
MOG, both coefficients refitted	0.729

The relation scores 0.651 here with its published a0 and 0.669 in Section 3.3 with a0 refitted on the disks. Here the Plummer scale puts exactly half the mass inside r_1/2, where Section 4.2 takes the scale as R_e, which puts 0.512 inside; an independent integration with that scale and its own folds agrees with the table within 0.01 dex.

With their published coefficients the kernels miss by factors near 70. Refitted on the dwarfs, all 108 fits drive the range parameter μ to its upper bound, collapsing

the kernel to a point and the theory to a constant multiplier of about 59; even then they do worse than the radial acceleration relation. Earlier studies found the same tension by other routes: no common MOG parameter pair for eight classical dwarfs with plausible stellar masses (Haghi and Amiri, 2016), and tensions in the nonlocal kernel's parameters in full Jeans analyses (De Martino, Della Monica and Roshan, 2025). The bound shows why. The extra pull a fixed linear kernel can give is capped by the matter it is given, and dwarfs have too little matter for their pull.

The bound covers isolated sources and constant coefficients. Running-parameter MOG, external fields, nonlinear responses and full covariant theories need their own treatment.

5. One acceleration scale does not carry across

5.1 The scale moves with the population

Fitted on the disks point by point, the radial acceleration relation takes a0 = 1.16 x 10^-10 m/s^2. Frozen at that value and carried to the other populations, its median acceleration errors are 50 percent for classical dwarfs, 86 percent for ultrafaint dwarfs, 34 percent for early-type galaxies, 48 percent for cluster centers and 69 percent for CLASH clusters. Refitted on all five populations together, a0 becomes 5.34 x 10^-10 and the disks lose their accuracy. Fitted on the X-COP clusters alone, it becomes 7.16 x 10^-10, six times the disk value. A law with one constant that needs a different constant for each kind of system is no longer one law. That clusters need more than MOND supplies is known (Tian et al., 2020; Famaey, Pizzuti and Saltas, 2025); the numbers here put the disk-to-cluster gap on one footing with the rest.

Freer formulas do not repair it. A six-coefficient quadratic in acceleration and radius wins when all populations are in training, 0.236 dex on held-out groups, and falls to 0.488 dex when a whole population is held out. Flexibility fits the populations it has seen.

5.2 Extra mass that follows hot gas, and not cold

Famaey, Pizzuti and Saltas (2025) proposed repairing MOND in clusters with an extra density that follows the hot gas, $\rho_{extra} = \eta\, \rho_{hot}\, e^{(-r/R_c)}$. With a_0 fixed at 1.2×10^{-10} and only η and R_c fitted, the repair transfers between cluster surveys: trained on 16 CLASH clusters ($\eta = 11.8$, $R_c = 289$ kpc), it predicts the seven X-COP clusters with a median error of 7.6 percent, against 53 percent for a single refitted a_0. The reverse transfer gives 17 against 38 percent, though within CLASH alone the repair does no better than one refitted a_0.

The repair is stated for hot gas. Carrying it to gas-rich disks tests whether it can become a rule for gas as such, whatever its temperature. In 61 disks where gas supplies at least half the baryonic pull, the fit prefers η near zero, with an 11.6 percent median speed error; carrying the cluster value over raises the speed error to 99.7 percent. Forced to share η and R_c, disks and clusters both deteriorate, and the joint fit pushes R_c to its upper bound. The extra mass appears where the gas is hot, inside deep cluster potentials, and not where it is cold, which is how a component tied to structure formation behaves and how a rule attached to gas does not.

5.3 A published halo relation at cluster scale

The halo side can be carried across the same gap, as an exploratory extension that rests on the published Moster relation alone; the paper's own three-coefficient rule remains a disk-to-dwarf result. The Moster relation of

Sections 3.2 to 3.5, unchanged, maps a central galaxy's stellar mass to its host halo's mass up to about 10^15 suns. A fifth protocol, hashed before any prediction was computed, applies it with nothing fitted to the 50 MaNGA brightest cluster galaxies and the 20 CLASH clusters of Section 2.1. Each brightest galaxy's published stellar mass and redshift set its halo, the concentration follows Dutton and Macciò (2014) at that redshift, and the measured baryons are added (Table 6); the CLASH stellar masses are those of Cooke et al. (2016), which Tian et al. take as total.

Table 6. Cluster-scale errors in dex with nothing fitted to these systems: the published Moster relation, inverted at its mean (fifth protocol) or with its scatter and the halo mass function (sixth protocol), MOND's relation and the disk-calibrated stellar rule.

Model	All 70 systems	MaNGA BCGs	CLASH centers	CLASH, 100 to 600 kpc
Published Moster relation, mean	0.185	0.182	0.154	0.204
Published Moster relation, with scatter	0.288	0.240	0.214	0.422
Radial acceleration relation, disk a0	0.403	0.338	0.479	0.550
Radial acceleration relation, published a0	0.388	0.327	0.460	0.526
Stellar rule, disk-calibrated	0.474	0.328	0.435	0.791

Inverted at its mean, the Moster relation cuts the error of MOND's relation by more than half, 0.185 against 0.403 dex, and the bootstrap interval of the gain, -0.248 to -0.188 dex, meets the protocol's test. MOND's relation falls short in every group: by 0.30 dex in the MaNGA galaxies, 0.45 dex at CLASH centers and 0.54 dex at 100 to 600 kpc. The disk-calibrated stellar rule fails at 100 to 600 kpc (0.79 dex), as Section 3.7 expects, since it has no bend above the Milky Way's mass.

Two audits, a sixth and a seventh protocol, each hashed before it ran, limit the result (Note S6). The relation is nearly flat at the top, so the mean inversion maps five of the 20 CLASH galaxies to halos above 10^16 suns, beyond any known cluster. Inverted instead with its published scatter, the measurement errors and the halo mass function of Tinker et al. (2008), the relation puts every halo's posterior median below 10^15 suns and still beats MOND's relation, 0.288 against 0.403 dex, a 29 percent cut rather than 54 (Table 6). Against the published lensing masses of the CLASH clusters (Umetsu et al., 2016), the mean inversion's masses scatter by 0.73 dex, a factor of five, and the probabilistic masses fall a median 0.98 dex low, as expected when the mass function's prior ignores how CLASH chose its clusters. The accelerations survive these checks; the halo masses do not. The test shows that halos set by starlight predict cluster-scale accelerations better than MOND's relation, not that the relation predicts each cluster's mass. Section 6.1 puts each sample's selection into the prior.

6. One formula for every scale

6.1 The formula

Sections 3 and 5.3 leave two halo descriptions: the stellar rule, set on disks, which carries to dwarfs, and the published Moster relation, which carries to brightest cluster galaxies. The rule fails at cluster scale because, above about 10^11 suns, the halos of gas-poor central galaxies grow far faster with stellar mass than those of disks (Section 3.8). As an exploratory extension, the two can be joined in one formula, with a switch set by a property each galaxy reports, its cold gas:

M_ref = M_p/f_b + 10^A (M_star/10^10 M_sun)^B (R_e/3 kpc)^C + q M_E(M_star, z),

q = (M_star/M_p)^50 / [1 + (M_t/M_star)^4],

with every halo held above 10^8 suns, M_halo = [M_ref^8 + (10^8 M_sun)^8]^(1/8), the scale below which gas could not cool to form the first galaxies (Johnson, 2013). The first two terms are the rule of Section 3.1, still set on disks alone. For galaxies chosen by their light, q is near one only for galaxies with almost no cold gas (M_p close to M_star) and stellar masses above a threshold M_t, the one further coefficient. In the final formula, M_t is set outside the benchmark, on the lensing halo masses of red central galaxies (Mandelbaum et al., 2016); in the intermediate rows of Table 7, it is fitted jointly on all populations at a weight of 0.1 against the disks. For the central galaxy of a sample chosen by its halos, q = 1, since that galaxy sits in such a halo by construction. Gas-rich and small galaxies keep the rule, and satellite dwarfs carry the subhalo factor of Section 3.3.

In its final form the formula also gives a core to the halos of diffuse gas-rich galaxies. For a system chosen by its light, let g = M_gas/M_p be the gas share of its baryons and L the logarithm of how far its stellar surface density, M_star/(2π R_e^2), lies below that of 10^10 suns with a half-light radius of 3 kpc, or zero if it lies above. Then the halo's reference mass M_ref is multiplied by 10^(t g L), and the dark mass inside radius r becomes

M_dark(r) = M_NFW(r) × tanh(r / 1.75 R_e)^n, n = tanh(κ g L),

where M_NFW(r) is the mass that the uncored halo holds inside r. The profile is the coreNFW of Read, Agertz and Collins (2016), with their core radius of 1.75 half-light radii. Its strength n, which they set by the time over which stars formed, is set here by gas and diffuseness, with two fitted numbers, t and κ. Systems without cold gas, dense ones and halo-selected ones are untouched. Fitted on all

149 disks, t = 0.150 and κ = 0.623.

Above the threshold the halo is M_E, the mean halo mass of galaxies of stellar mass M_star at redshift z in the sample at hand: the halo mass function of Tinker et al. (2008) times a log-normal distribution of stellar mass about the Moster relation with its redshift evolution, 0.18 dex wide (its 0.15 dex scatter and 0.10 dex for stellar-mass errors), times the sample's selection. For samples chosen by their light the selection is one. For samples chosen by their halos it keeps the halos the survey selected: clusters above about 5 × 10^14 suns for CLASH, chosen for X-ray temperatures above 5 keV (Postman et al., 2012), and halos above 10^14 suns for the MaNGA brightest cluster galaxies, the stated aim of the program that targeted them (SDSS Collaboration, 2020). Without scatter or selection, M_E is Section 5.3's mean inversion. Nothing in M_E is fitted.

Two published corrections enter without new coefficients. Massive early-type galaxies have higher stellar mass-to-light ratios than a Chabrier initial mass function gives, rising with velocity dispersion: log α = 0.38 log10(σ_e/200 km/s) - 0.06 relative to a Salpeter function, with 0.12 dex scatter (Posacki et al., 2015). The stellar pull of each galaxy with a measured dispersion, the 50 MaNGA galaxies and the 16 early types (Cappellari et al., 2013), is raised by α at that dispersion, converted to the Chabrier scale (Chabrier masses divided by 0.61 give Salpeter masses; Madau and Dickinson, 2014). And the concentration relation of Section 3.1 describes relaxed halos (Dutton and Macciò, 2014), while a lensing stack averages over all halos of a given mass, so the stacks' concentrations are multiplied by the ratio of the median concentrations of all and of relaxed halos in the Millennium Simulation, (4.67/5.26)(M_200 h/10^14 M_sun)^-0.01, about 0.93 at 10^12 suns (Neto et al., 2007). Rotation-curve samples, chosen for regular kinematics, and CLASH, chosen for relaxed X-ray morphology, keep the relaxed relation.

6.2 Built in steps, scored held out

The test reuses the five populations and adds the four KiDS stellar-mass stacks of Section 3.8, inside 0.3 Mpc. Every score comes from 60 partitions: three assignments of the disks to five folds, each crossed with four held-out lensing stacks. In each partition, A, B and C are refitted on the training disks, M_t on the training systems of every population (in the last three rows, on the red centrals), t and κ, in the last row, on the training disks, and the held-out systems are scored. The tenth to sixteenth protocols, the tenth in several rounds, and the nineteenth, which adds the core, were each hashed before their runs, but the formula was reached over several rounds on these data, so Table 7 is a retrospective test; the blind test below is not.

Table 7. Held-out errors in dex and the lensing chi-square summed over the four held-out stacks. Each row adds one step to the row above it. The stellar rule and the seventh and eighth rows have three coefficients fitted to the benchmark, the last row five, and the other rows four.

Model	Disks	Dwarfs	Early types	MaNGA	CLASH centers	CLASH, 100 to 600 kpc	Lensing
Stellar rule (Section 3.1)	0.153	0.406	0.203	0.322	0.386	0.774	79.0
One formula: switch, mean Moster halo, floor	0.153	0.370	0.203	0.184	0.155	0.208	72.0
Plus the IMF trend of spheroids	0.153	0.370	0.185	0.114	0.155	0.209	71.0
Plus the expected halo with each sample's selection	0.153	0.370	0.185	0.118	0.105	0.136	69.5
Plus the switch on for halo-selected samples	0.153	0.370	0.185	0.116	0.106	0.132	69.5

Model	Disks	Dwarfs	Early types	MaNGA	CLASH centers	CLASH, 100 to 600 kpc	Lensing
Plus all-halo concentrations in lensing stacks	0.153	0.370	0.186	0.116	0.106	0.132	64.5
Plus the threshold set on SDSS red centrals	0.153	0.370	0.185	0.116	0.106	0.132	64.7
Plus hot gas among the clusters' baryons	0.153	0.370	0.185	0.116	0.102	0.100	64.7
Plus cores in diffuse gas-rich disks	0.153	0.370	0.185	0.116	0.102	0.100	64.5

Against the rule alone, the final formula (last row) keeps the disks, lowers the lensing chi-square from 79.0 to 64.5, and cuts the errors by 0.67 dex for the clusters at 100 to 600 kpc, 0.28 dex at their centers and 0.21 dex for MaNGA, with host-bootstrap intervals of the gain of -0.71 to -0.64, -0.35 to -0.22 and -0.23 to -0.18 dex. The subhalo factor and the floor improve the dwarfs by 0.036 dex (-0.074 to -0.000).

Each later row adds one step (Note S7 gives each in full). The IMF trend, with nothing refitted to MaNGA, removes the 0.15 dex by which the MaNGA galaxies sat above the formula. The expected halo with each sample's selection lowers the cluster errors from 0.209 to 0.136 dex at 100 to 600 kpc and from 0.155 to 0.105 at the centers, and with no selection the same expectation comes within 0.14 dex rms of the lensing halo masses of the red SDSS centrals of Mandelbaum et al. (2016), against 0.50 for the mean inversion: at fixed stellar mass, cluster centrals sit in the heavy tail of the halo distribution, and galaxies chosen by their light near its mean. The all-halo concentrations lower the lensing chi-square from 69.5 to 64.5, with a size that comes from Neto et al. (2007), not from a fit. Setting the threshold on the red centrals takes it off the benchmark: it comes out at 10^11.33 suns, from 10^11.31 to 10^11.41 over the 60 partitions, and every held-out score

stays within 0.4 percent, so that only the rule's three coefficients, set on disks, are then fitted to the benchmark and every other population is predicted. Counting the clusters' hot gas among their baryons, with nothing fitted, lowers the error at 100 to 600 kpc from 0.132 to 0.100 dex (-0.045 to -0.020). The core changes the benchmark little, the disks from 0.1533 to 0.1531 dex and the lensing chi-square from 64.7 to 64.5; its gain is on the points the benchmark drops (Section 6.4), and with it five coefficients are fitted to the benchmark, all of them on disks. The lensing correction, the hot-gas step and the core were each chosen after a look at these data.

A blind test is less favorable to the brightest galaxy's role. A twelfth protocol, hashed before any value was read, scored the formula of the fourth row on the 216 HeCS-omnibus clusters with a measured brightest-galaxy stellar mass (Sohn et al., 2020), none used here, with the selection their authors describe, dynamical masses above about 10^14 suns. Against the caustic masses M_200, the median offset is -0.11 dex, but the rms is 0.58 dex, larger than the 0.41 dex spread of the masses about their median. Above 10^11.4 suns the prediction is unbiased (-0.03 dex) and as good as the median (0.38 against 0.37 dex), but the brightest galaxy's stellar mass tracks its cluster's mass only weakly. At cluster scale the formula therefore predicts the typical mass of a selected sample, not the mass of each cluster. The fifth row, with the switch on for every central of a halo-selected sample, was written after this test; it brings the HeCS rms to 0.45 dex, but HeCS can no longer test it.

Four limits remain (Note S7). First, the threshold rests on massive central galaxies chosen by their light: set on the SDSS red centrals it is stable, but the formula follows their halo masses only to 0.13 dex. Second, the high-mass branch needs each sample's selection, and a stated cut is not the whole selection: with the hot gas counted, the formula's halo for a CLASH cluster is 1.0 × 10^15 suns,

still 0.20 dex below the clusters' lensing masses in the median, and its accelerations at 100 to 600 kpc fall short by 0.07 dex on average, most of their 0.10 dex error. Third, for massive star-forming galaxies the halos are too heavy. From the Milky Way's published stars and gas (McMillan, 2017), the formula predicts its rotation curve within 7 percent of the Gaia-based curve (Eilers et al., 2019), but with a halo of 1.8 × 10^12 suns, against 1.37 × 10^12 in McMillan's fit. Fourth, the KiDS profiles still prefer concentrations below the relaxed relation by more than the all-halo correction supplies, and the final formula gives a quarter to a third of the measured early-late split. Most of the remaining lensing chi-square sits in the outermost bin of each stack, next to the 0.3 Mpc beyond which Brouwer et al. (2021) find that satellites missed by the isolation criterion raise the signal, and in one negative point of the lightest stack.

6.3 Against MOND and standard halos at every scale

The final formula can be set against the two standard accounts on one footing: the same systems, radii, weights and lensing covariance, with nothing fitted to the benchmark, so that their errors on all data are already held out (Table 8). MOND is the radial acceleration relation with its published constant, a0 = 1.2 × 10^-10 m/s^2 (McGaugh, Lelli and Schombert, 2016), without the external field, which only worsens its dwarfs (Section 3.5); its lensing is its phantom mass, projected, on the assumption, as in Brouwer et al. (2021), that lensing measures the same acceleration as dynamics. Standard halos are those of the published Moster relation, as the expected halo of each sample with its selection (Section 6.1), with the same concentrations, the same all-halo correction for the lensing stacks and the same floor. Each account is given its better case: both take the IMF trend of spheroids, which lowers their errors on the early types and MaNGA, and the standard halos leave out the subhalo factor, which raises their dwarf error, and

take the same baryon term for the clusters as the formula (Note S8 gives the numbers). The comparison was made after the formula was frozen, and nothing in it is fitted. It is a comparison on the benchmark's points; Section 6.4 extends it to the rotation-curve points that the benchmark's cut on velocity errors removes and to galaxies outside the benchmark.

Table 8. The final formula against MOND's relation and standard halos on the benchmark of Table 7. Errors in dex, held out for the formula and the stellar rule, on all data for the two accounts, which fit nothing to the benchmark; lensing chi-square summed over the four stacks, on the stricter mask in parentheses. MOND and standard halos take the IMF trend of spheroids, and standard halos leave out the subhalo factor and take the formula's baryon term for the clusters.

Model	Disks	Dwarfs	Early types	MaNGA	CLASH centers	CLASH, 100 to 600 kpc	Lensing
Final formula (Table 7, last row)	0.153	0.370	0.185	0.116	0.102	0.100	64.5 (39.4)
Stellar rule (Section 3.1)	0.153	0.406	0.203	0.322	0.386	0.774	79.0 (54.1)
MOND's relation	0.155	0.633	0.182	0.186	0.460	0.526	89.0 (51.3)
Standard halos (Moster)	0.166	0.440	0.182	0.116	0.102	0.101	103.6 (82.1)

The formula is never significantly worse than either account. Against MOND, the host-bootstrap interval of the difference is -0.011 to +0.008 dex on disks (with a0 refitted, MOND scores 0.153, Section 3.2) and -0.005 to +0.010 on early types, and the formula is lower on dwarfs by 0.26 dex (-0.35 to -0.17), on MaNGA by 0.07 (-0.09 to -0.05) and on the clusters by 0.36 and 0.43 dex (-0.43 to -0.28 and -0.45 to -0.40), with a lensing chi-square lower by 24 (12 on the stricter mask). Against standard halos, it is lower on disks by 0.012 dex (-0.023 to -0.002) and on

dwarfs by 0.07 (-0.12 to -0.03), equal on early types (-0.008 to +0.014) and, by construction, on MaNGA and the clusters, whose halos it takes from the same expected halo and baryon term, and lower in lensing chi-square by 39 (43 on the stricter mask). Each account fails where its premise does. MOND's single acceleration scale fails in clusters, where its relation gives about a third of the measured pull (0.43 and 0.52 dex too little on average at the centers and at 100 to 600 kpc). Standard halos fail where a galaxy's halo depends on more than its stellar mass, in disks and dwarfs, whose sizes and gas the rule reads. The formula keeps each where it works.

In size, the formula's errors are those of the measurements on every dynamical scale, for the 310 systems of the benchmark: 149 disks, 75 dwarfs, 16 early types, 50 MaNGA galaxies and 20 CLASH clusters, with stellar masses from 5 × 10^2 to 1.5 × 10^12 suns (Table 9). With the formula's prediction taken as the truth and each system's inputs redrawn from their published errors 2,000 times, measurement alone produces a benchmark error of 0.144 dex for the disks and 0.38 dex for the dwarfs (Note S8 lists the errors redrawn). The formula's held-out errors all lie inside the central 95 percent of the draws, the early types at the upper edge. By the size of its errors, these data cannot tell the formula from a perfect one. The test is blunt, and the formula is not alone in passing it. MOND's relation passes on the disks and the early types and fails on the dwarfs, MaNGA and the clusters. Standard halos, given the better case above, pass on every dynamical scale and fail in lensing; the paired comparisons above, which the formula wins on disks and dwarfs, are the sharper test between the two. In lensing the formula's own chi-square, 64.5 for 29 points, is too high for noise. On the 24 points left when each stack's outermost bin and the one negative point are set aside (Section 6.6) it is 21.3, against 37.2 for MOND's relation and 68.7 for standard halos. The six dynamical scales and the lensing on the stricter mask,

which was fixed before the formula was built, make seven tests. The formula's lowest share among them is 3.4 percent, and a perfect formula would show a share that low in at least one of seven tests one time in five. At a level of 5 percent over the seven together, 0.7 percent each, the formula passes all seven, MOND's relation fails five and standard halos fail the lensing. On the primary mask all three fail.

Table 9. The formula's held-out errors against measurement noise alone. Noise: the benchmark error when the formula's prediction is taken as the truth and each system's inputs are redrawn from their published errors, 2,000 times; for lensing, the chi-square distribution for the number of points. Last three columns: the percentage of draws at least as large as the error of the formula, of MOND's relation and of standard halos (Table 8). Errors in dex; lensing in chi-square, the 24 points on all data with the frozen coefficients.

Scale	Formula	Noise, median	Noise, 95 percent range	Formula, percent	MOND, percent	Standard halos, percent
Disks	0.153	0.144	0.122 to 0.176	24	22	8
Dwarfs	0.370	0.383	0.217 to 0.574	55	0.4	26
Early types	0.185	0.146	0.106 to 0.191	4	5	5
MaNGA	0.116	0.112	0.088 to 0.143	39	under 0.1	39
CLASH centers	0.102	0.134	0.093 to 0.193	93	under 0.1	93
CLASH, 100 to 600 kpc	0.100	0.088	0.070 to 0.111	15	under 0.1	13
Lensing, primary mask, 29 points	64.5	28.3	16.0 to 45.7	0.02	under 0.01	under 0.01
Lensing, stricter mask, 25 points	39.4	24.3	13.1 to 40.6	3.4	0.15	under 0.01
Lensing, 24 points	21.3	23.3	12.4 to 39.4	62	4	under 0.01

Patterns within the residuals test more sharply than their size, and five exceed the noise. First, within single disks the rotation curves depart from the formula by

0.066 dex where the velocity errors give 0.045 (0.043 to 0.048); MOND's relation and standard halos leave more within disks, 0.072 and 0.080 dex, and from disk to disk the formula's offsets are those of the errors (0.134 dex against 0.136), with no trend against stellar mass, size, gas fraction, surface brightness, inclination or distance. Second, at fixed stellar mass the dwarfs' residuals fall with half-light radius by 0.50 dex per dex, where the redrawn samples give -0.27 to +0.28; the tenth of the dwarfs most extended for their mass and the tenth most compact carry the trend. Third, the MaNGA residuals fall with baryonic acceleration (correlation -0.59 against -0.48 to +0.17) and rise with half-light radius (+0.55); most of this lies in the published inputs (Section 6.6). Fourth, the CLASH clusters at 100 to 600 kpc lie 0.07 dex above the formula on average, where noise gives 0.02 either way. Fifth, the lensing points set aside lie off the formula together: each stack's outermost bin above it, and the innermost point of the lowest-mass stack, a negative excess surface density, four standard errors below it. None of the five is fitted here.

6.4 Every rotation-curve point: beyond the quality cut

The benchmark keeps only rotation-curve points with velocity errors of at most 10 percent, the criterion of McGaugh, Lelli and Schombert (2016). The formula's core (Section 6.1) was added for the points this removes. With it the formula matches or beats MOND's relation and standard halos on those points too, held out and on galaxies outside the benchmark (Table 10).

Table 10. SPARC disks beyond the benchmark's cut on velocity errors, and galaxies outside the benchmark. Host-balanced errors in dex. Held out: each benchmark disk is scored with coefficients fitted without it, in the partitions of Table 7. Outside: every coefficient is frozen at its value from the 149 benchmark disks. MOND: its relation with the published constant, and with the constant refitted on the same training points and weights as the core's two numbers.

Set	Galaxies	Points	Formula	Without its core	MOND	MOND, constant refitted	Standard halos
Benchmark points, held out	149	2,700	0.153	0.153	0.155	0.153	0.166
Dropped points, held out	114	438	0.344	0.401	0.364	0.352	0.407
All points, held out	149	3,138	0.193	0.218	0.200	0.195	0.229
SPARC galaxies outside the benchmark, all points	26	251	0.392	0.435	0.415	0.396	0.453
Dwarf irregulars outside SPARC, all points	8	86	0.308	0.411	0.414	0.392	0.459

The cut removes 582 points in 139 of SPARC's 175 galaxies, most of them inside the half-light radius. On them all three accounts predict accelerations too high, and without the core the formula does worse than MOND's relation: 0.416 dex, against 0.381 for MOND's relation and 0.426 for standard halos (formula without the core minus MOND's relation, +0.023 to +0.047). The difference lies in the 201 points that rotate slower than 30 km/s; on the faster points the three tie, at 0.323, 0.326 and 0.326. The tie with MOND's relation on disks (Section 3.2) therefore held for points measured to 10 percent: in the slowly rotating centers of small disks, cuspy halos predict too much. Pressure support does not explain the excess (Note S9).

Two terms aimed at this failed their protocols, a seventeenth and an eighteenth, each hashed before its held-out run (Note S9). A fitted weight on cold gas in the baryon term raised the lensing chi-square from 64.7 to 75.7, outside its tolerance. Larger, looser halos for diffuse gas-rich disks, with the halo mass multiplied by 10^(t g L) and the concentration by 10^(-s g L), lowered the held-out disk error from 0.1533 to 0.1513 dex, short of the protocol's mark, a gain beyond 1 percent with a paired bootstrap range below zero: the range ran from -0.005 to +0.002.

The core keeps a mass term like that one and replaces the lower concentration by a cored profile (Section 6.1). Its protocol, the nineteenth, set three marks before the held-out run: every benchmark score within its tolerance of the eighth row's in Table 7; and, on the held-out disks' dropped points and on all their points, a gain over the formula without the core with a paired bootstrap range below zero, and an error below that of MOND's relation. All were met. On the dropped points of the held-out disks the error falls from 0.401 to 0.344 dex (-0.077 to -0.039), below the 0.364 of MOND's relation (-0.031 to -0.009) and the 0.407 of standard halos (-0.087 to -0.042); below 30 km/s it is 0.42, against 0.55 without the core and 0.46 for MOND's relation. Over all points of the held-out disks, with no cut, it is 0.193 dex, against 0.218 without the core (-0.038 to -0.014), 0.200 for MOND's relation (-0.015 to +0.002) and 0.229 for standard halos (-0.051 to -0.022). The formula thus beats MOND's relation on the dropped points and matches it over all points, and beats standard halos on both. MOND's relation is taken here, as everywhere in this paper, with its published constant. Refitted on the same training points with the same weights, the constant falls to 0.99 × 10^-10 m/s^2 and MOND's errors to 0.352 dex on the dropped points and 0.195 over all points; the formula stays lower on both, but the paired ranges then include zero (-0.018 to +0.001 and -0.009 to +0.005).

Two outside checks, forecast in the protocol with every coefficient frozen and every point kept, were also met. Iorio et al. (2017) give circular velocities, corrected for asymmetric drift, for 17 LITTLE THINGS dwarf irregulars. Eight of the ten outside SPARC have 3.6 micrometer fluxes in the Local Volume Legacy survey (Dale et al., 2009) and stellar scale lengths in Read et al. (2017), from which stellar masses and sizes were built on SPARC's scale. On those eight the formula's error is 0.308 dex, against 0.411 without the core and 0.414 for MOND's relation (-0.181 to -0.032 against MOND's relation; lower in six of the eight). On the 26 SPARC galaxies outside the benchmark, excluded for quality or inclination or left without a benchmark point, it is 0.392, against 0.435 and 0.415 (-0.036 to -0.007). Against MOND's relation with the refitted constant the first check still holds and the second becomes a match.

The limits of this test were fixed in the protocol. It is retrospective: the dropped points were first scored for this paper, and about a dozen core forms were looked at on all data before the protocol. And the dropped points remain the one place where no account reaches the measurement errors. If the formula were exact, the published errors would give 0.273 dex on the 582 points (0.235 to 0.352). With its coefficients from all 149 disks the formula gives 0.360, a result that 2 percent of such draws reach; MOND's relation and standard halos, at 0.381 and 0.426, lie further out, at 1 and 0.5 percent.

6.5 What physics the formula stands for

Each piece of the formula has a counterpart in galaxy formation with cold, collisionless dark matter. Three physical alternatives tested here fit worse, and three kinds of dark-matter particle do not do what its core does.

The stellar term follows abundance matching for disks: for a disk of typical size and gas, the rule's halo lies within 0.25 dex of the expected halo of the Moster

relation from 10^8 to 5 × 10^10 suns. Above that it keeps rising as a single power of stellar mass, without the break of abundance matching, as rotation-curve fits to SPARC also find (Posti, Fraternali and Marasco, 2019). The switch is the other side of the same split: at fixed stellar mass, red central galaxies sit in halos more than twice as massive as blue ones (Mandelbaum et al., 2016). The size term has the sign of simulated galaxy formation, in which galaxies are smaller in more concentrated halos (Jiang et al., 2019; Sun, Jiang and Wang, 2026). The regularity that MOND's relation describes emerges in cold-dark-matter simulations, with a scatter of 0.08 dex (Ludlow et al., 2017). The core has its counterpart too. In simulations, gas that star formation drives in and out of a dwarf's center turns its cusp into a core about the size of the stellar half-light radius, the more so the longer stars keep forming (Read, Agertz and Collins, 2016). The formula's cores sit where that picture puts them, in diffuse disks that still hold their gas, and are absent from dwarf spheroidals and cluster centers, which need their density high (Note S10).

Three readings that go beyond cold dark matter were tested on all data, on the benchmark's points, and none improves the fit there. Self-interacting dark matter, with the isothermal cores of Kaplinghat, Tulin and Yu (2016) and Ren et al. (2019) and a cross-section falling with speed from about 2 cm^2/g in galaxies to 0.1 cm^2/g in clusters, raises the formula's dwarf error from 0.364 to 0.518 dex and its disk error from 0.150 to 0.167; held out on those points, the training disks choose no self-interaction in all 60 partitions. Halo contraction at full strength (Cautun et al., 2020) raises the standard halos' disk error from 0.166 to 0.175 dex, and concentrations set by galaxy size with the simulated relation of Jiang et al. (2019), applied directly, raise it to 0.25.

The core raises the question again. Three kinds of dark-matter particle also change the inside of a halo, each by a rule that follows the halo and not the gas,

and each with one number that must be the same in every halo: fuzzy dark matter, whose solitons follow a core-halo relation (Schive et al., 2014); warm dark matter, which lowers the concentrations of small halos (Schneider et al., 2012); and self-interacting dark matter, with the isothermal cores above. Under a twentieth protocol, hashed before its held-out run, each took the core's place in the formula. Its number was set on all points of the training disks with the core's weights and carried unchanged to every population, and the marks were the core's three (Section 6.4). None meets them. The training disks choose no fuzziness in any of the 60 partitions. They choose a warm thermal relic of 2.0 to 2.4 keV, which more than doubles the dwarf error, from 0.370 to 0.769; counts of Milky Way satellites exclude relics below 6.5 keV on their own (Nadler et al., 2021).

On all points, unlike on the benchmark's, the training disks do choose self-interaction: 1 cm^2/g at low speed in 13 of 15 disk partitions. It lowers the held-out error on the dropped points by 14 percent (-0.082 to -0.033), as the core does, but it raises the error on the benchmark's own points by 5.7 percent (+0.003 to +0.015), on the dwarfs by 25 percent (+0.036 to +0.150) and at the cluster centers by 26 percent (-0.004 to +0.056). Self-interaction makes the cores that the slow centers of gas-rich disks need, and it puts cores in halos that do not want them; the core set by gas and diffuseness appears only where it is needed. On this benchmark the cores follow the gas and not the halo. The three recipes are the published ones for dark matter alone, so this is evidence, not proof, that the particle is cold and collisionless and that the baryons make the cores.

Whether the gate's gas stands in for a history of star formation was tested under a twenty-first protocol, hashed before any score was computed. Among the disks, which all still form stars, the amount of gas does not matter: with the gate changed to diffuseness alone in any system that holds cold gas, and the core's

two numbers refitted, every held-out score is level or slightly lower (0.189 against 0.193 dex over all points; -0.009 to +0.001). In gas-free dwarfs a history gate and the gas gate differ, and there the data lean to history without deciding. For the eight dwarf spheroidals whose dark-matter density at 150 pc was measured by Read, Walker and Steger (2019), the formula's cusps give an error of 0.21 dex and the published history recipe of Read, Agertz and Collins (2016), with nothing fitted, 0.13; Fornax, which formed stars for 12 Gyr, has a third of the cusp's density, as the history recipe predicts. With eight galaxies the paired range of the difference includes zero (-0.23 to +0.06). The same measurements show a limit of the core: in eight dwarf irregulars the formula with its core predicts a density at 150 pc twice the measured one, so the core is in the right galaxies and too shallow at the very center. The formula keeps the gas fraction, which every sample provides; a gate built on history needs a history for every gas-free dwarf, and 20 of the 75 have one (Weisz et al., 2015).

6.6 What remains, and the data that would test it

By the size of its errors the formula is at the limit of the data on every dynamical scale (Table 9), and in lensing once two known problems are set aside: without each stack's outermost bin and the one negative point, its chi-square on all data is 21.3 for 24 points. What the data can still test lies in the patterns.

Seven things remain (Note S11). One, in the MaNGA galaxies, lies mostly in the published inputs; the others cannot be repaired with the data in hand without fitting the test itself. Within single disks the curves still depart from the formula by more than their velocity errors, though by less than from MOND's relation and standard halos (Section 6.3), and on the points that the benchmark's cut removes no account reaches the measurement errors (Section 6.4). The outermost lensing bins lie above the formula, next to the radius beyond which missed satellites raise

the signal (Brouwer et al., 2021). Extended dwarfs fall below the formula and compact ones above it, and with the orbits in hand tides do not look like the cause: the slope with radius is -0.43 among the 16 Milky Way satellites with pericenters beyond 50 kpc (Pace, Erkal and Li, 2022) and -0.56 among the 20 inside. Dense MaNGA galaxies fall below it and extended ones above. Nothing tested within the formula removes this; rebuilding their published accelerations from one consistent light profile lowers their error to 0.087 dex and the trend with it, with nothing fitted (below). The CLASH clusters stay 0.07 dex above it on average, because their stated cut understates how they were chosen. The halos of Milky-Way-size spirals come out too heavy. And blind, the formula gives a cluster sample's typical mass, not each cluster's.

Ten forecasts, each written and hashed before any test of it, name the data that would decide these points (Table 11). The first six were written for the formula before its core was added; the core changes none of the quantities they name by as much as 1 percent. The tenth is the one the formula is most likely to fail: the formula gives a gas-free dwarf a cusp whatever its history, and Fornax already lies low.

Table 11. The ten forecasts, each hashed before any test of it. Note S11 gives each in full.

Forecast	Data	The formula holds if	It fails if
1	Isolated lenses chosen with spectroscopic redshifts, as in the Galaxy And Mass Assembly survey, scored as here	The chi-square per point falls below 1.5 and the outermost bins no longer lie above the formula	The value is above 2 with the excess still there
2	Dwarfs with dispersions measured to better than 10 percent, pericenters beyond 50 kpc and no tidal features	The residuals' slope with radius at fixed stellar mass lies between -0.2 and +0.2 dex per dex, against -0.5 now	The slope is steeper than -0.3

Forecast	Data	The formula holds if	It fails if
3	Clusters chosen by a stated threshold in mass, with their hot gas and brightest galaxies measured	The accelerations at 100 to 600 kpc come out within 0.05 dex of the formula on average, with an error below 0.13 dex	The offset is beyond 0.10 or the error above 0.16
4	Star-forming central galaxies of 3 to 10 × 10^10 suns, with halo masses from lensing or satellite motions	A predicted failure: the halo masses lie 0.1 to 0.3 dex below the formula's, as for the Milky Way	No failing mark was set
5	Independent, deeper sizes and stellar masses for the 50 MaNGA galaxies	The correlation of their residuals with radius falls below 0.3	It stays above 0.45. Tested: failed as worded
6	The 50 MaNGA galaxies' accelerations rebuilt from single-Sérsic fits of images at least one magnitude deeper than SDSS's	The error is below 0.10 dex and the correlation below 0.3	The error is above 0.116 or the correlation above 0.45
7, 8	At least ten gas-rich dwarf irregular or low-surface-brightness disks in neither SPARC nor the sample of Iorio et al., with rotation curves corrected for asymmetric drift and every point kept	With its core the formula has a lower error than without it, with a range that excludes zero, and a lower error than MOND's relation (8); the looser-halo term, not adopted, lowers the error as well (7)	The core raises the error, or lowers it but not below that of MOND's relation
9	At least ten gas-poor dwarfs in none of this paper's samples, each with a dynamical mass at or inside its half-light radius	The formula has a lower error than the same halos with self-interaction of 1 cm^2/g, with a range that excludes zero	The self-interacting halos have the lower error
10	At least four gas-free dwarfs outside the eight of Read, Walker and Steger (2019) whose stars formed over at least 9 Gyr, each with a dark-matter density measured inside 0.3 half-light radii	The measured density lies within 0.2 dex of the formula's on average	It lies lower by 0.2 dex or more, as in Fornax; the gate must then be rebuilt on the history of star formation

The fifth forecast has been tested, with the Sérsic fits of the NASA-Sloan Atlas (Blanton et al., 2011) in the MaNGA catalog of the seventeenth SDSS data

release (Abdurro'uf et al., 2022). As worded, with only the stellar accelerations rebuilt, the forecast fails: the residuals' correlation with radius is 0.51 (95 percent bootstrap range 0.21 to 0.72), above the 0.45 that counts against it. But Tian et al. derive the measured accelerations from the same light profile, through the Jeans equation, and the forecast left that out. Rebuilt from one Sérsic profile throughout, for light, mass and size, the 50 galaxies give the formula an error of 0.087 dex instead of 0.116 (95 percent range of the difference -0.053 to -0.008), and the correlation with radius falls to 0.28 (-0.04 to 0.55). Standard halos improve alike, to 0.088 dex, and MOND's relation stays at 0.17. This full rebuild was chosen after the worded test failed, so the sixth forecast commits to it on deeper images. If deeper images agree with the atlas, the published accelerations, not the formula, carry the MaNGA pattern.

7. Why cluster benchmarks at overdensity radii are circular

Scoring acceleration relations in clusters at overdensity radii such as R_500 or R_200 builds in a circularity that can pass for unification. In the X-COP clusters, a shared acceleration-and-radius formula scored at R_500 shows a 5.63 percent median error on held-out clusters, and the accuracy comes from the definition of R_500. That radius encloses 500 times the critical density ρ_crit = 3H(z)^2/(8πG), so M_500 = (4π/3) 500 ρ_crit R_500^3, which fixes the target acceleration at g_500 = G M_500/R_500^2 = 250 H(z)^2 R_500. A formula in radius and redshift alone reproduces the target to 0.0001 percent, with no baryonic information. Holding out whole clusters does not break an identity built into the radius. The clusters here are therefore scored at fixed physical radii, and any cluster benchmark scored at R_500 or R_200 should be checked for the same identity. The circularity concerns tests that predict the acceleration at a radius built from the same mass; measuring a cluster's mass at R_500 is not circular.

8. What it means

The results point the same way. Halos whose mass follows the stars carry from disks to dwarfs and, joined with a published halo relation in one formula, to clusters and lensing, with no rotation data from the predicted galaxy. A modification that turns ordinary matter into extra effective matter through a fixed kernel cannot reach what most dwarfs need. One acceleration scale, or one rule for extra mass attached to gas, fitted where it works, fails when carried to the next kind of system. The missing mass behaves like matter: it has its own amount, set by how the galaxy formed and traced by its stars, and gravity acts on it as on everything else.

Description length, counted roughly in coefficients rather than bits, gives the same ranking from another side: a description earns its weight by what it predicts per bit. A halo rule of three coefficients that carries from disks to dwarfs is shorter, for what it predicts, than a force law that needs a new constant or a new extra component for each population. The results fit a broader pattern: short laws hold with no built-in scale, as $\nabla^2\Phi = 4\pi G\rho$ does, and the variety lives in the state the laws produce over time. A force law that builds a scale into itself, through a kernel's range or an acceleration threshold, puts the complexity in the law, and here such laws failed when carried to systems of different density and history. A dark component that obeys the ordinary law gathers where baryons cool in its potential well, which would explain why a galaxy's stars and size predict its unseen mass. A search for a short fundamental rule, or for a hidden sector, can take these numbers as targets: whatever the dark component is, it must collect around stars in the proportions found here.

The comparison has limits. It fits no raw lensing shear, cluster temperature, cosmic microwave background or large-scale structure likelihood; a complete case for either side must, and relativistic MOND theories already compute

cosmological spectra (Skordis and Zlosnik, 2021). MOND's external-field effect enters only through the estimate of McGaugh and Milgrom (2013), which takes the host's field as uniform. Dwarf masses rest on equilibrium estimators. The halos use a median concentration relation and no formation histories. Each protocol was fixed before its run on data inspected in earlier runs, and the formula was reached over several rounds on these data. Its blind test recovers only typical cluster masses (Note S12).

A fit cannot prove what dark matter is. Proving it would take detecting the dark component directly, or a rule that reproduces every population to measured precision. Ideally the rule would also predict something new that is then confirmed. Section 3.6 makes such predictions and fixes them with a hash.

The next tests follow. Spectra of the nine Local Group dwarfs of Table 4 check the forecasts directly, and tighter measurements of Grus II and Andromeda XII will show whether the rule's two misses are scatter.

Lensing adds another. Brouwer et al. (2021) find that, at the same stellar mass, early types show lensing accelerations 0.17 dex (split by Sérsic index) to 0.27 dex (by color) above those of late types, and note that a gravity modification independent of other galaxy properties could not explain this. The stellar rule's size term gives a quarter to a third of the split (Section 3.8). Two tests follow: halo mass against size at fixed stellar mass and type, where the rule predicts lighter halos for larger disks, and a halo whose concentration, rather than mass, depends on size, fitted to rotation curves and scored on lensing. A modified gravity law that hopes to cover all scales must pass the mass budget of Section 4 and one constant across disks, dwarfs and clusters.

9. Conclusion

One formula for the missing mass of galaxies, with five numbers fitted on disk galaxies and one threshold set on published lensing masses, predicts the accelerations of 310 galaxies and clusters to within their measurement errors, and matches or beats both MOND and standard dark-matter halos at every scale tested, from dwarf galaxies to clusters and lensing, and over all rotation-curve points. Its backbone, a three-coefficient halo rule driven by stellar mass and size, matches MOND's radial acceleration relation on unseen disks and predicts dwarf galaxies 28 percent better, and 25 dwarfs outside the original sample 45 percent better. Joined with the published Moster relation, it cuts the rule's cluster error from 0.77 to 0.10 dex, though a blind test on 216 clusters recovers their typical mass but not each cluster's. A core in the halos of diffuse gas-rich disks, two of the formula's five numbers, takes it 5 percent below MOND's relation on the rotation-curve points that the usual quality cut removes, and none of three kinds of dark-matter particle, fuzzy, warm or self-interacting, does what the core does: the cores follow the gas, not the halo. Fixed kernels of nonlocal gravity and MOG with constant parameters cannot exceed a factor $1 + 2\alpha$ at a source's half-mass radius, and most dwarfs need more. MOND's fitted acceleration scale grows six times from disks to clusters, and its hot-gas cluster repair cannot be carried to cold-gas disks. What remains is stated with the data that would settle it: five patterns within the residuals, a lensing profile flatter than predicted, and whether the cores follow the gas or the history of star formation, against ten hashed forecasts and fixed predictions for 264 dwarfs not yet measured. In these tests, halos set by starlight carry across and the universal force laws do not. The law is the same everywhere, and the variety lives in the matter it acts on.

Data availability

The code, the frozen inputs, selection and exclusion tables with provenance and hashes, all coefficients, the prediction tables and saved results for every fold, the twenty-one protocols and ten forecasts with their hashes, independent numerical checks and the scripts that reproduce every number here are public at https://github.com/MatheismOrg/papers, in the folder SUPPLEMENTS/A19_Reproduce (commit 37d6421 of 11 November 2026), with the original analysis branches beside it in SUPPLEMENTS/A19_Original_Analysis. The twelve Supplemental Notes give the full account of every test summarized here, and Note S12 lists what the repository holds. The repository records two coding errors in its own checks, found by review after the runs and fixed; neither changes a number reported here. It credits the SPARC, LITTLE THINGS, Local Volume Legacy, nearby-galaxy, X-COP, CLASH, MaNGA, NASA-Sloan Atlas, KiDS, HeCS-omnibus and stellar-history catalogs on which it rests. This work collects no new observations.

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